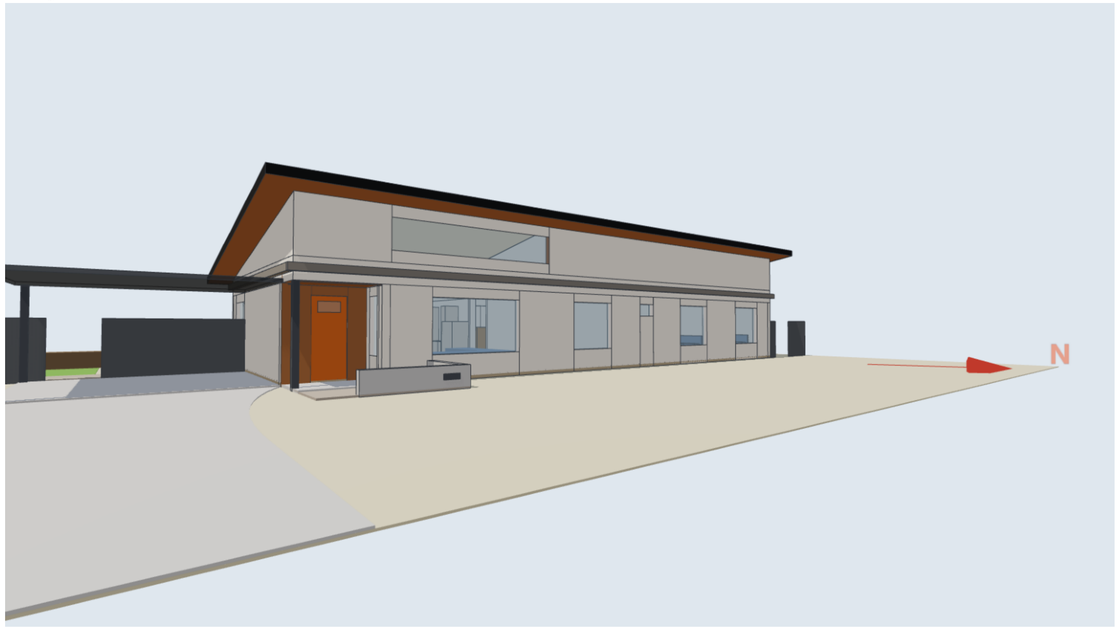
ACOSTA RESIDENCE

New single-family residence, Eaton Fire rebuild · 220 Laurel Dr, Altadena CA 91001

3D massing, from the street (for reference)



SCOPE OF WORK
New one-story, all-electric single-family residence (4 bedrooms, 3 baths) with an attached open carport, replacing the house destroyed in the Eaton Fire. Demolition of the remaining retaining wall and chimney, grading and drainage per civil, new driveway, utilities (water, sewer, electric; no gas). Fire sprinklers throughout.

PROJECT TEAM
Owner: Richard Acosta
Plans prepared by: _____
Structural engineer: _____
Civil engineer: _____
Energy consultant: _____
Fire sprinkler designer: _____

GENERAL NOTES
1. Work shall comply with the codes listed and the conditions of approval.
2. Do not scale drawings; dimensions govern. Verify on site.
3. Structural, civil and energy documents govern their own scope.
4. Special inspections and structural observation per the structural engineer.
5. Smoke and CO alarms per A-2; fire sprinklers per the deferred submittal.

SHEET INDEX

SHEET	TITLE
A-0	Cover sheet, code analysis
A-1	Site plan, project data
A-2	Floor plan, door and window schedules
A-3	Roof plan
A-4	Exterior elevations
A-5	Building sections
A-6	Fire-zone details (CRC R337)
A-7	Attic and structural loads plan
S-	Structural (by the structural engineer)
C-	Grading and drainage (by the civil engineer)
T-24	Energy compliance (CF1R) and solar PV (by the energy consultant)
CALGreen	Green building checklist

DEFERRED SUBMITTALS

#	ITEM
1	Fire sprinkler plans and calculations (NFPA 13D), before the final foundation inspection
2	Roof truss drawings and calculations (truss manufacturer), reviewed by the engineer of record
3	Solar PV plans, if not in the energy package

CODE ANALYSIS

ITEM	PROJECT
Codes	2025 California Residential, Building, Electrical, Mechanical, Plumbing, Energy, Fire and Green Building Standards Codes; LA County Title 26 amendments
Occupancy	R-3 single-family dwelling; U open carport (attached)
Construction	Type V-B; automatic fire sprinklers throughout (NFPA 13D)
Stories / height	1 story; ~17'-6" above finished floor (30' limit, Altadena CSD)
Areas	Dwelling 2,238 sq ft · porch 29 · carport 324 sq ft
Lot / coverage	7,319 sq ft · 2,592 sq ft (35.4%)
Zoning	R-1-7500 · Altadena CSD · Disaster Recovery Permit (non-like-for-like)
Fire zone	VHFHSZ: wildland-urban interface construction, CRC R337 (A-6)
Energy	All-electric; Title 24 Part 6 compliance by the energy consultant
Green building	CALGreen mandatory measures (checklist)

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard
Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer
(separate sheets)

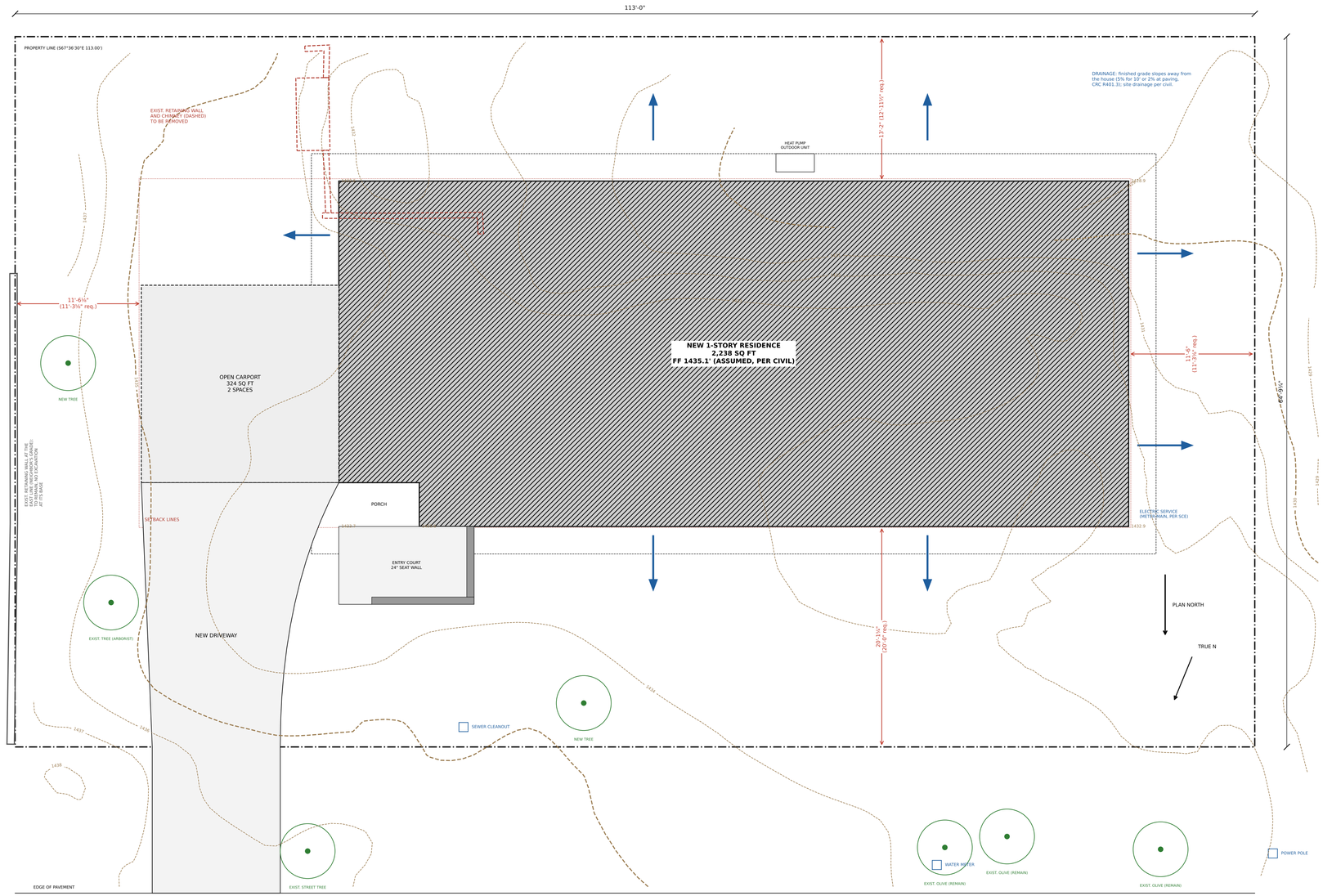
DRAFT FOR REVIEW
not for construction

Date: October 15, 2026
Revisions:

© 2026 Richard Acosta.
All rights reserved.
These drawings are for
220 Laurel Dr only. Not to
be copied, adapted, sold
or built from on any
other site.

COVER SHEET, CODE ANALYSIS

A-0



LAUREL DRIVE (50' R/W)

SITE PLAN 3/16" = 1'-0"

PROJECT DATA

Scope	New 1-story single-family residence with attached open carport; Eaton Fire rebuild (non-like-for-like). Demolish remaining retaining wall and chimney.
Address / APN	220 Laurel Dr, Altadena CA 91001 APN 5832-003-027
Zoning	R-1-7500 · Altadena CSD
Fire zone	Very High Fire Hazard Severity Zone
Occupancy	R-3 (dwelling), U (carport)
Construction type	V-B, fully sprinklered (NFPA 13D)
Codes	2025 California Residential, Building, Electrical, Mechanical, Plumbing, Energy and Green Building Codes, as amended by LA County Title 26
Lot	113.00' x 64.77' = 7,319 sq ft
Floor area	2,238 sq ft (to outside of walls)
Porch / carport	29 / 324 sq ft
Lot coverage	2,592 sq ft (35.4%); limit 2,830
Height	1 story, ~17'-6" above finished floor
Bedrooms / baths	4 / 3
Parking	2 covered (carport)

SITE NOTES

1. Lot lines, existing trees, utilities, spot elevations and 1' contours (dashed) from the owner's topographic survey.
2. Finished floor elevation, grading and drainage per civil (separate sheets).
3. Eaves (dashed) 2'-6" all round (LACC 22.110.090).
4. Existing street and parkway trees protected during construction (root zones).
5. 5' noncombustible zone around the house.
6. Water, sewer and SCE service per the utilities; no gas service (all-electric).
7. Demolition: remove the remaining retaining wall and chimney from the burned house (dashed red); asbestos survey first if required (SCAQMD Rule 1403); fill and grade per civil.
8. The retaining wall along the east line holds the neighbor's higher grade: protect it in place.

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard
Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer
(separate sheets)

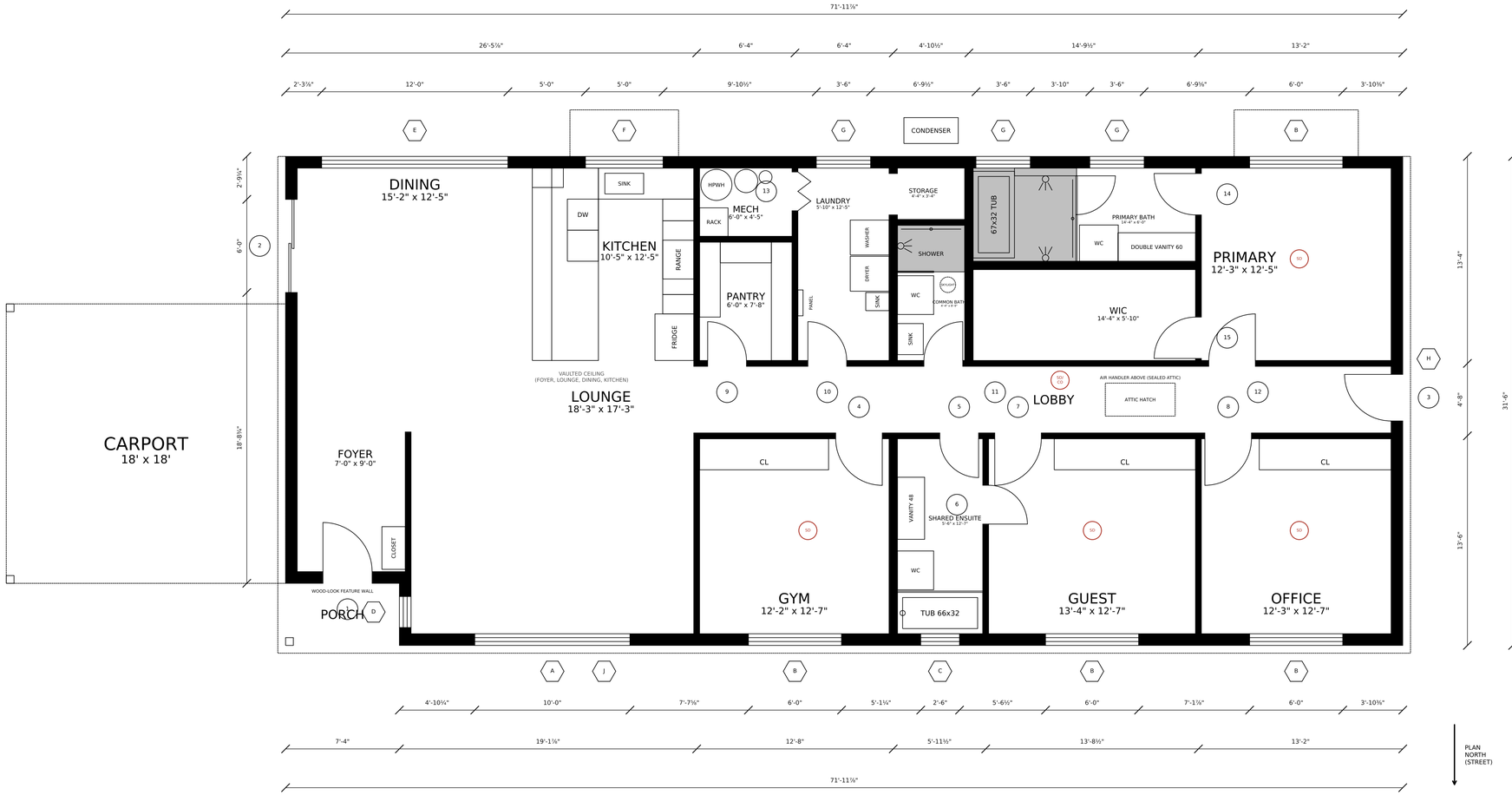
**DRAFT FOR REVIEW
not for construction**

Date: October 15, 2026
Revisions:

© 2026 Richard Acosta.
All rights reserved.
These drawings are for
220 Laurel Dr only. Not to
be copied, adapted, sold
or built from on any
other site.

**SITE PLAN
& PROJECT DATA**

A-1



FLOOR PLAN 1/4" = 1'-0"

GENERAL NOTES

1. Dimensions are to wall centrelines; exterior walls to the outside face of framing + finish (9" walls).
2. Walls: exterior 2x6 with 1" rigid insulation and stucco; interior 2x4; plumbing walls 2x6.
3. Ceilings 9'-0" unless noted. The foyer, lounge, dining and kitchen are vaulted under the roof (see sections).
4. Doors 8'-0" high unless noted; interior doors solid core. Exterior doors and glazing per CRC R337 (VHFHSZ).
5. All exterior glazing dual-pane with at least one tempered pane (CRC R337.8.2.1). Fully tempered where noted (CRC R308.4).
6. Bedroom windows meet emergency escape: 5.7 sq ft net clear, 20" min width, 24" min height, sill 44" max (CRC R310).
7. Smoke alarms: hard-wired, interconnected, battery backup, at least 3' from bathroom doors (CRC R314). Combination smoke / CO alarms outside the sleeping areas.
8. All-electric: heat pump HVAC (air handler in the sealed attic; outdoor unit at the rear wall), heat pump water heater (HPWH) in the mechanical closet. Whole-house ventilation by a continuous exhaust fan.
9. Fire sprinklers throughout (NFPA 13D); plans by the sprinkler designer (deferred submittal).
10. Attic access in the hall: 22" x 54" pull-down ladder or insulated access panel, 30" min headroom above (CRC R807.1).
11. Furniture not shown; closets, cabinets and fixtures shown for reference.

DOOR SCHEDULE

TAG	LOCATION	SIZE (W x H)	TYPE	NOTES
1	PORCH / FOYER (exterior)	3'-0" x 8'-0"	Hinged	Solid core 1-3/8"+ or noncombustible (CRC R337.8.3); weatherstripped; glass lite, if any, dual-pane tempered
2	DINING (exterior)	6'-0" x 8'-0"	Sliding glass door	Dual-pane tempered glazing (CRC R308.4.1, R337.8.2); weatherstripped
3	LOBBY (exterior)	3'-0" x 7'-0"	Hinged	Solid core 1-3/8"+ or noncombustible (CRC R337.8.3); weatherstripped; solid slab, no glass; fixed transom above
4	GYM / LOBBY	3'-0" x 8'-0"	Hinged	
5	SHARED ENSUITE / LOBBY	2'-6" x 8'-0"	Hinged	
6	SHARED ENSUITE / GUEST	2'-6" x 8'-0"	Hinged	
7	GUEST / LOBBY	3'-0" x 8'-0"	Hinged	
8	OFFICE / LOBBY	3'-0" x 8'-0"	Hinged	
9	LOBBY / PANTRY	2'-6" x 8'-0"	Hinged	
10	LOBBY / LAUNDRY	2'-6" x 8'-0"	Hinged	
11	LOBBY / COMMON BATH	2'-6" x 8'-0"	Hinged	
12	LOBBY / PRIMARY	3'-0" x 8'-0"	Hinged	
13	MECH / LAUNDRY	2'-6" x 8'-0"	Bifold, louvered	louvered (water heater air)
14	PRIMARY BATH / PRIMARY	2'-8" x 8'-0"	Hinged	
15	WIC / PRIMARY	2'-8" x 8'-0"	Hinged	

WINDOW SCHEDULE

TAG	QTY	LOCATION	SIZE (W x H)	SILL	OPERATION	NOTES
A	1	Lounge	10'-0" x 5'-6"	2'-6"	Fixed center, slider each end	
B	4	Gym, Guest, Office, Primary	6'-0" x 5'-6"	2'-6"	Horizontal slider	Emergency escape (note 6)
C	1	Shared Ensuite	2'-6" x 1'-6"	6'-6"	Awning	Obscure glass
D	1	Foyer	2'-0" x 5'-6"	2'-6"	Fixed, reeded glass	Tempered (CRC R308.4.2)
E	1	Dining	12'-0" x 6'-6"	1'-6"	Fixed, two units mulled	
F	1	Kitchen	5'-0" x 4'-6"	3'-6"	Horizontal slider	
G	3	Primary Bath, Laundry	3'-6" x 2'-0"	6'-0"	Awning	
H	1	Lobby	3'-0" x 1'-6"	7'-2"	Fixed transom	
J	1	Lounge	18'-0" x 3'-0"	12'-1½"	Fixed, 3 units mulled	Clerestory

LEGEND

- 1 — Door tag (see door schedule)
- A — Window tag (see window schedule)
- SD — Smoke alarm (SD), smoke / CO alarm (SD/CO)
- Overhead: band, canopy, attic hatch, skylight

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

DRAFT FOR REVIEW not for construction

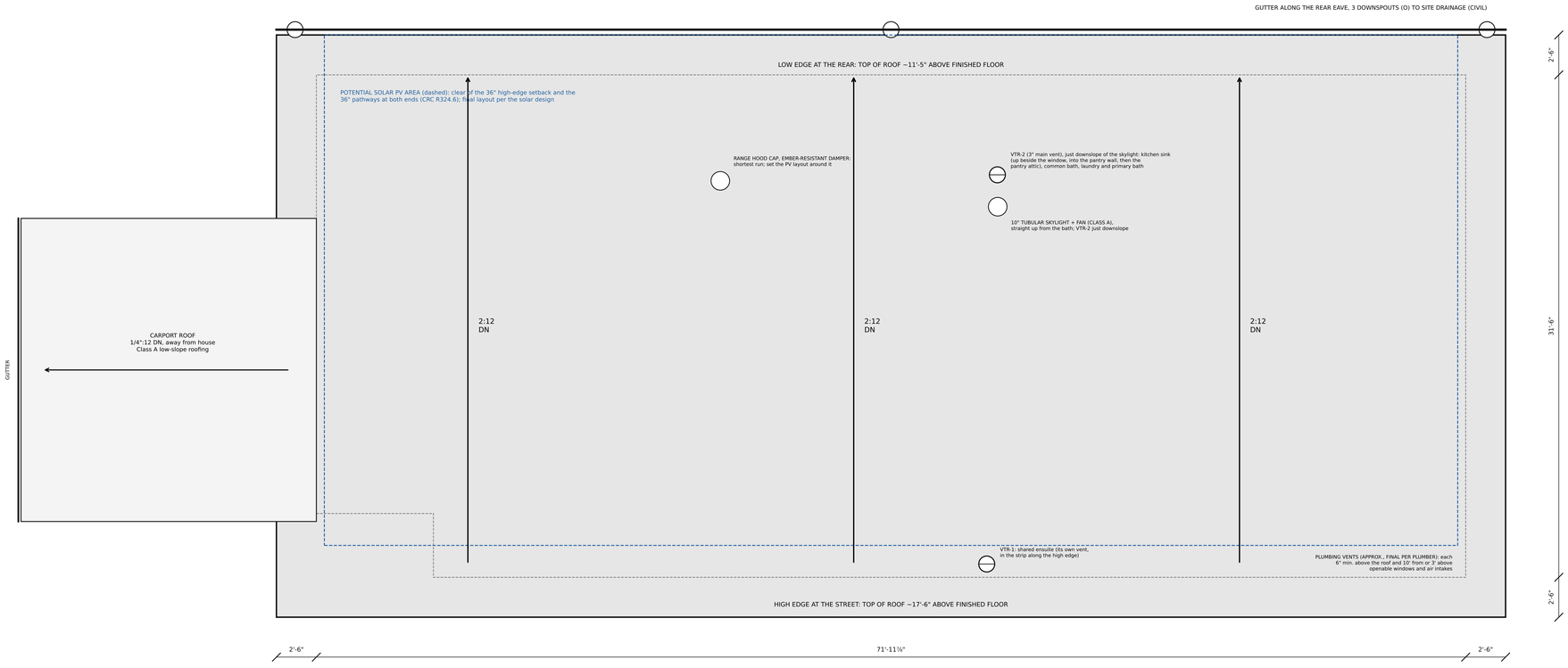
Date: October 15, 2026
Revisions:

© 2026 Richard Acosta. All rights reserved. These drawings are for 220 Laurel Dr only. Not to be copied, adapted, sold or built from on any other site.

FLOOR PLAN, DOOR & WINDOW SCHEDULES

A-2

ROOF NOTES
1. Main roof: single slope at 2:12, high at the street. Class A asphalt shingles over the low-slope double underlayment (CRC R905.2.2); shingle line warranted for 2:12 (dark, cool-rated per Title 24).
2. Sealed (unvented) attic, CRC R806.5: insulation at the roof deck; NO attic or eave vents.
3. Eaves 2'-6" all round, enclosed; dark metal fascia and drip edge. See A-6 for fire-zone details (CRC R337).
4. Gutters with noncombustible gutter guards (CRC R337.5.4); downspouts to the site drainage system per civil.
5. Roof framing, sheathing and connections per structural.
6. Roof penetrations flashed. Two plumbing vents (VTR): the shared ensuite's in the strip along the high edge; the rest combined into one beside the common bath skylight, so the panel layout works around one cluster. Bath and laundry fans exit through walls where possible. The range hood cap is the other penetration in the PV area.
7. Carport: attached open carport, its own low-slope roof (Class A), gutter on its east edge.
8. Future solar: layout and mounting by the solar provider. Its conduit enters through the roof into the attic and runs inside to the panel: none on the roof surface or the exterior walls.



STREET (PLAN NORTH)

ROOF PLAN 1/4" = 1'-0"

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

DRAFT FOR REVIEW not for construction

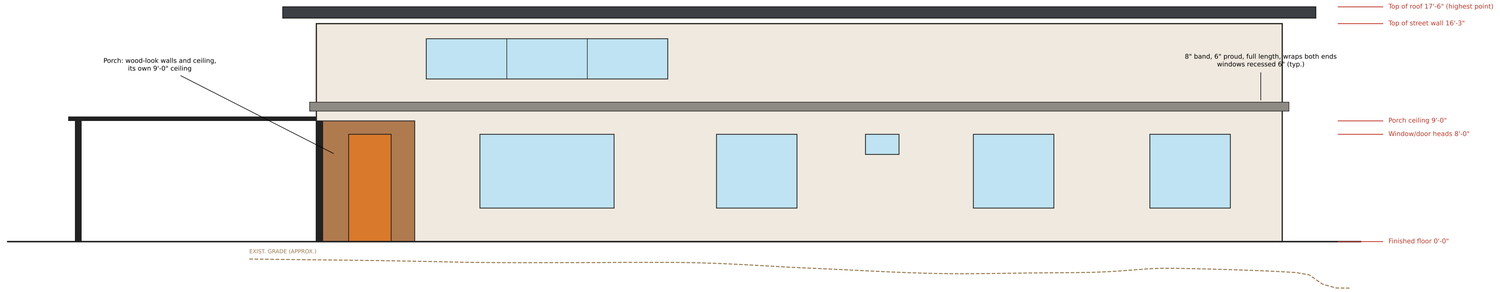
Date: October 15, 2026
Revisions:

© 2026 Richard Acosta. All rights reserved. These drawings are for 220 Laurel Dr only. Not to be copied, adapted, sold or built from on any other site.

ROOF PLAN

A-3

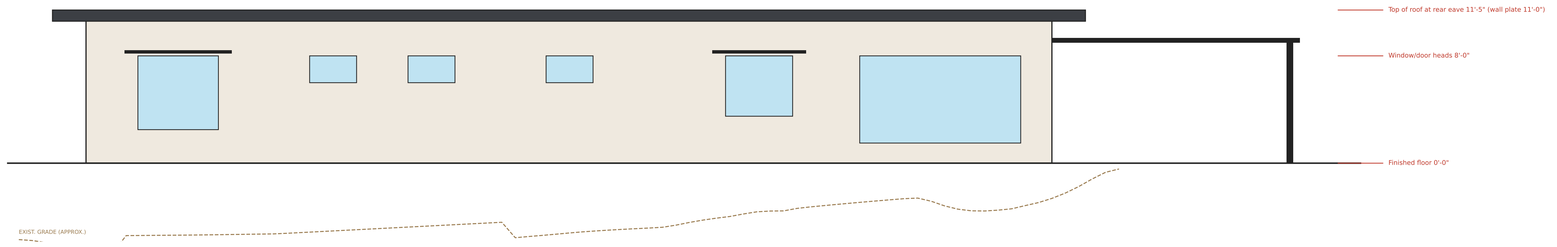
NORTH (STREET) ELEVATION



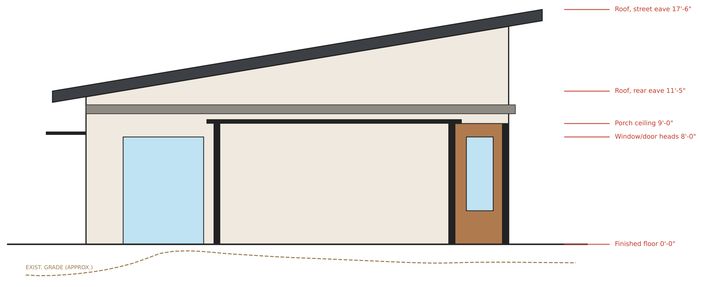
EXTERIOR MATERIALS
1. Walls: 3-coat stucco, light color (noncombustible).
2. Roof: Class A asphalt shingles, dark (A-3).
3. Fascia, drip edge, band and canopies: dark metal.
4. Street-wall band: 8" high, 6" proud, wraps both ends.
5. Porch walls and ceiling: wood-look fiber cement or porcelain (noncombustible / ignition-resistant).
6. Windows: dual-pane, one pane tempered (CRC R337.8); street windows recessed 6". See A-2 schedule.
7. Eave soffits enclosed, no vents (A-6).
8. Existing grade (dashed) is approximate, from the survey with an assumed finished floor; finished grades and floor elevation per civil.

Heights are above finished floor (0'-0").
Assumed finished floor: 1435.1' (6" above the highest existing grade at the walls).
Scale: 3/16" = 1'-0"

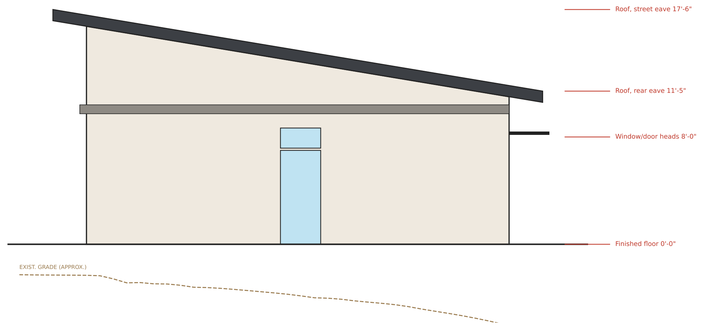
SOUTH (REAR) ELEVATION



EAST (CARPORT) ELEVATION



WEST ELEVATION



ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

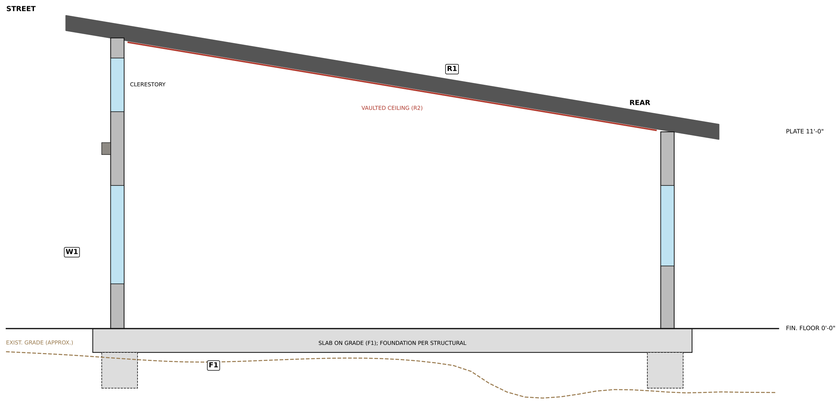
DRAFT FOR REVIEW
not for construction

Date: October 15, 2026
Revisions:

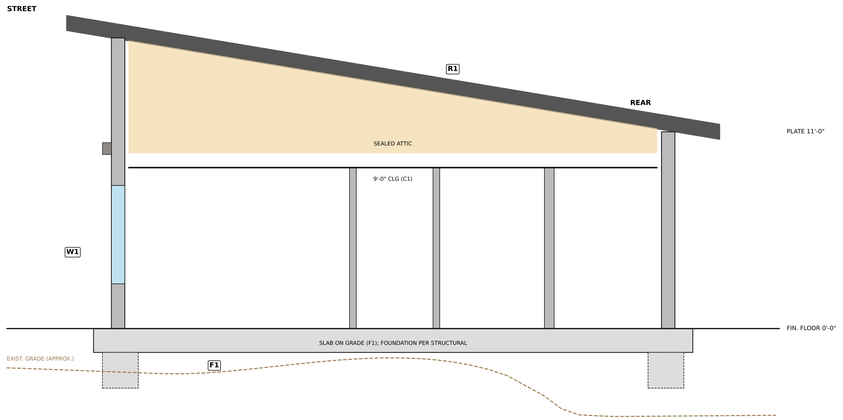
© 2026 Richard Acosta. All rights reserved. These drawings are for 220 Laurel Dr only. Not to be copied, adapted, sold or built from on any other site.

EXTERIOR ELEVATIONS

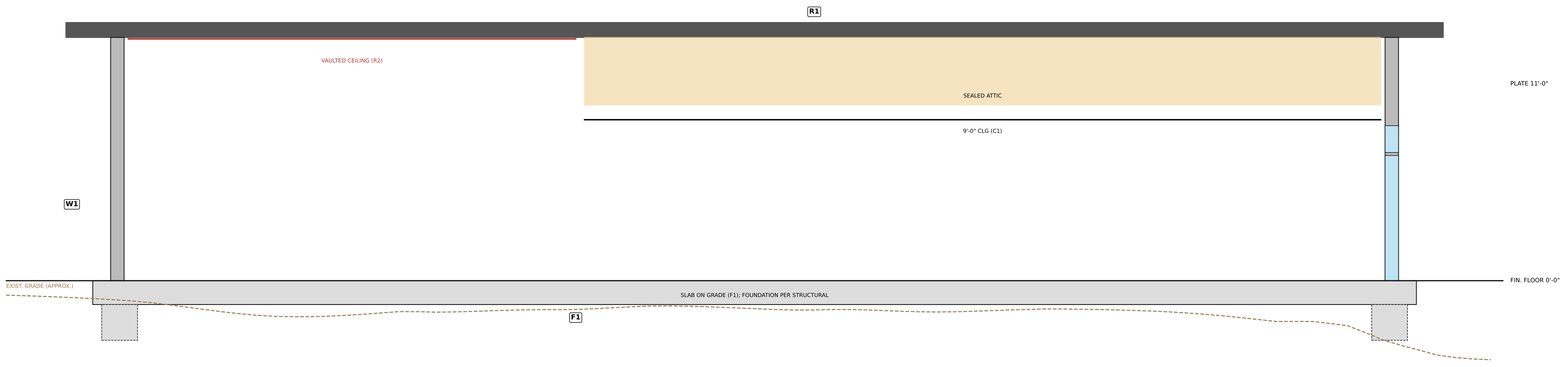
A-4



SECTION A: THROUGH THE LOUNGE AND KITCHEN (VAULTED) 1/4" = 1'-0"



SECTION B: THROUGH THE GUEST ROOM, LOBBY AND WALK-IN CLOSET 1/4" = 1'-0"



SECTION C: ALONG THE LOBBY, LOOKING SOUTH 1/4" = 1'-0"

ASSEMBLIES (insulation values per the Title 24 report)
R1 ROOF: Class A shingles, low-slope double underlayment, roof sheathing, framing per structural; spray foam at the underside of the deck (unvented, CRC R806.5); ignition barrier / thermal barrier where the attic is used.
R2 VAULT: as R1, gypsum board on the framing (no attic).
C1 CEILING (9'-0"): 5/8" gypsum board; sealed attic above.
W1 EXTERIOR WALL: 7/8" 3-coat stucco, weather barrier, 1" rigid insulation, sheathing, 2x6 studs with cavity insulation, 1/2" gypsum board.
W2 INTERIOR WALL: 2x4 (plumbing walls 2x6), gypsum both sides; sound insulation at bedrooms, baths, laundry.
F1 FLOOR: concrete slab on grade over vapor retarder and capillary break (CALGreen 4.505.2); foundation, slab and footings per structural; floor level per civil.

Section cuts: A at the lounge / kitchen, B at the guest room, C along the lobby (see A-2). Heights above finished floor; existing grade approximate (survey, assumed floor 1435.1').

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

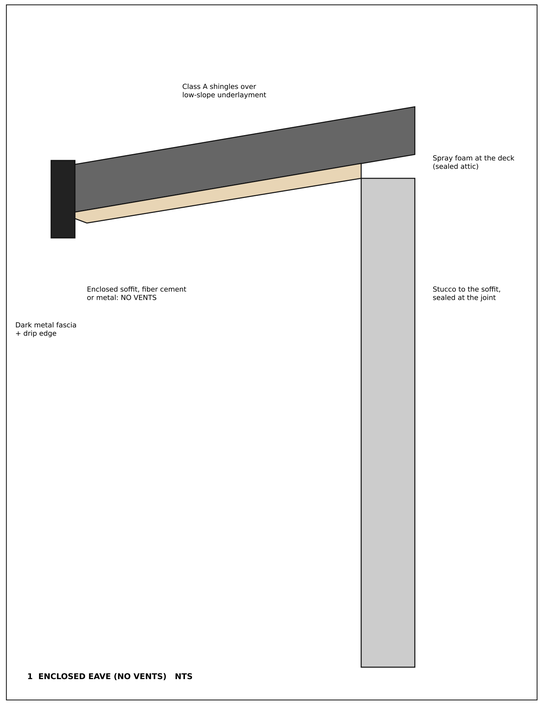
DRAFT FOR REVIEW
not for construction

Date: October 15, 2026
Revisions:

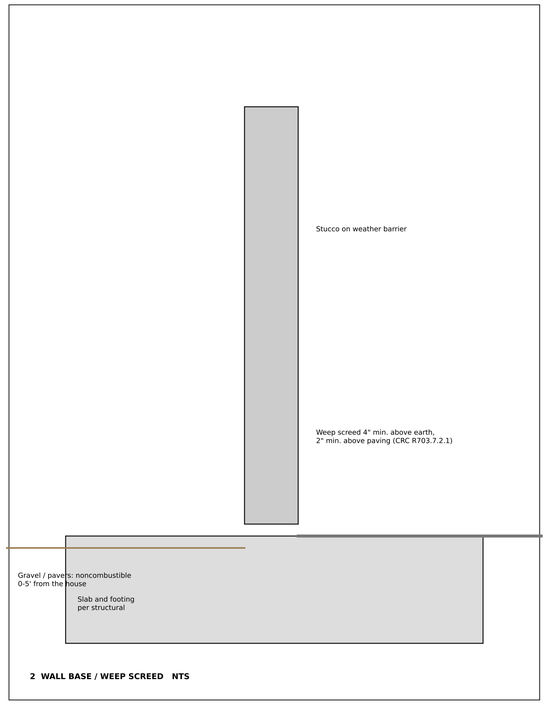
© 2026 Richard Acosta. All rights reserved. These drawings are for 220 Laurel Dr only. Not to be copied, adapted, sold or built from on any other site.

BUILDING SECTIONS

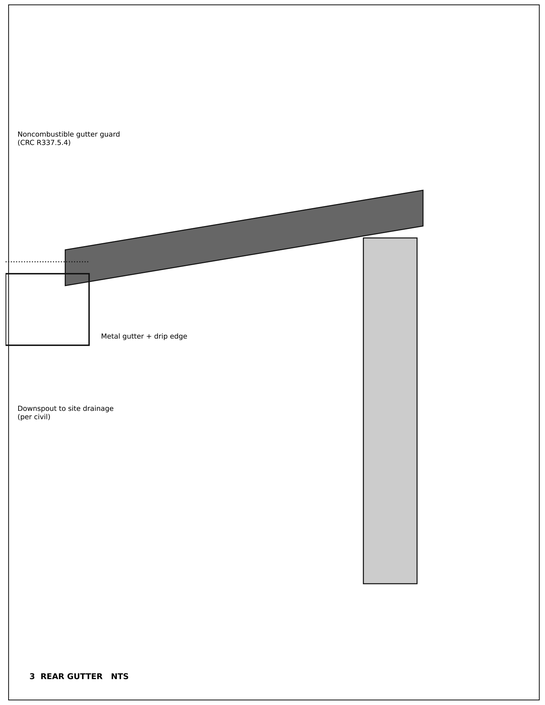
A-5



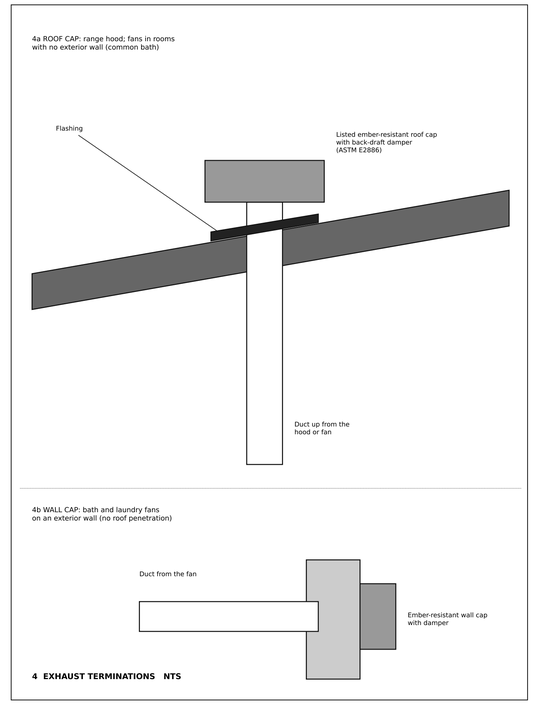
1 ENCLOSED EAVE (NO VENTS) NTS



2 WALL BASE / WEEP SCREED NTS



3 REAR GUTTER NTS



4 EXHAUST TERMINATIONS NTS

WILDLAND-URBAN INTERFACE REQUIREMENTS (VERY HIGH FIRE HAZARD SEVERITY ZONE)

CRC	ITEM	THIS PROJECT
R337.5	Roof	Class A asphalt shingles; low-slope double underlayment. Valleys and edges metal-flashed.
R337.5.4	Gutters	Noncombustible gutter guards on all gutters.
R337.6	Vents	No attic or eave vents (sealed attic). Exhaust and hood terminations: listed ember-resistant vents (ASTM E2886) or noncombustible 1/16"-1/8" mesh with dampers.
R337.7	Exterior walls	3-coat stucco (noncombustible). Porch cladding: noncombustible or ignition-resistant wood-look (fiber cement / porcelain).
R337.7.5	Eaves, soffits	Enclosed soffits, noncombustible or ignition-resistant (fiber cement or metal); fascia dark metal. Porch ceiling the same.
R337.8.2	Glazing	Dual-pane, at least one tempered pane, all windows, doors, skylight, clerestory.
R337.8.3	Exterior doors	Noncombustible, ignition-resistant, or solid core at least 1-3/8" thick.
R337.9	Decks	None. At-grade concrete landing and entry court only.
R337.10	Carport, fences	Carport roof Class A; framing within 5' of the house noncombustible or ignition-resistant. Fences and gates within 5' of the house noncombustible.
--	Zone 0 (0-5')	Noncombustible ground cover (gravel, pavers, concrete) within 5' of the house; no bark mulch or plants against the walls.

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard
Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer
(separate sheets)

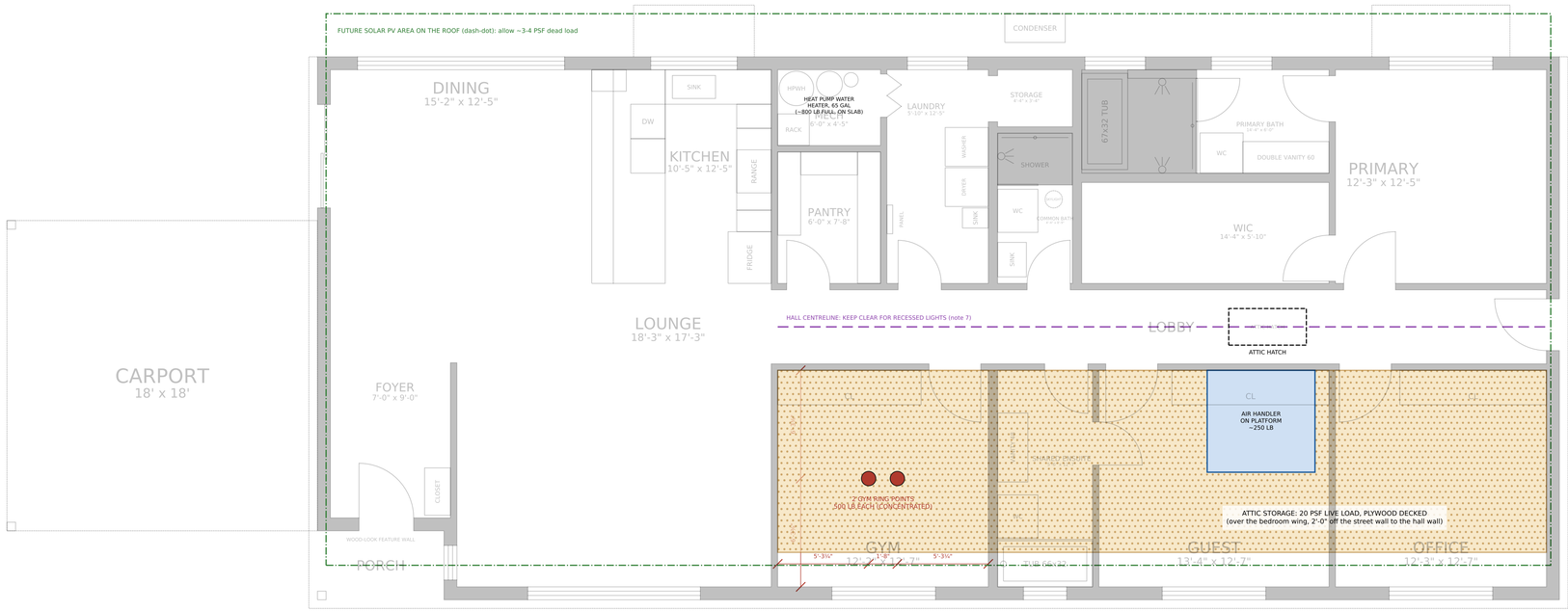
**DRAFT FOR REVIEW
not for construction**

Date: October 15, 2026
Revisions:

© 2026 Richard Acosta.
All rights reserved.
These drawings are for
220 Laurel Dr only. Not to
be copied, adapted, sold
or built from on any
other site.

**FIRE-ZONE
DETAILS
(CRC R337)**

A-6



ATTIC AND LOADS PLAN 1/4" = 1'-0"

LOAD NOTES (for the structural engineer and truss designer)

1. Gym ring points: two eye bolts 20" apart at the gym's centre, rings swinging along the room's long (street-to-hall) direction. 500 lb concentrated at each, dynamic: blocking across several trusses or joists with framing connectors; through-bolted forged eye bolts with plate washers.
2. Attic storage: 20 psf live load on the decked strip shown (hatched), over the bedroom wing where the shed roof gives the attic its height; light storage only.
3. Air handler: on a platform beside the attic hatch with a 30" walkway, light and outlet (mechanical code).
4. Future solar: on the roof plane, clear of the fire setbacks (A-3); include its dead load in the roof design.
5. Heat pump water heater: on the slab in the mechanical closet.
6. Vaulted great room: no attic and no loads above the ceiling.
7. Bedroom hall ceiling: keep the hall's centreline free of blocking, strongbacks, ducts and pipes, so recessed lights fit between the trusses along it (clear of the hatch, smoke alarm and return grille). Run the duct trunk to one side.

ACOSTA RESIDENCE

Single-family residence
fire rebuild (Eaton Fire)

220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

DRAFT FOR REVIEW not for construction

Date: October 15, 2026
Revisions:

© 2026 Richard Acosta. All rights reserved. These drawings are for 220 Laurel Dr only. Not to be copied, adapted, sold or built from on any other site.

ATTIC AND STRUCTURAL LOADS PLAN

A-7